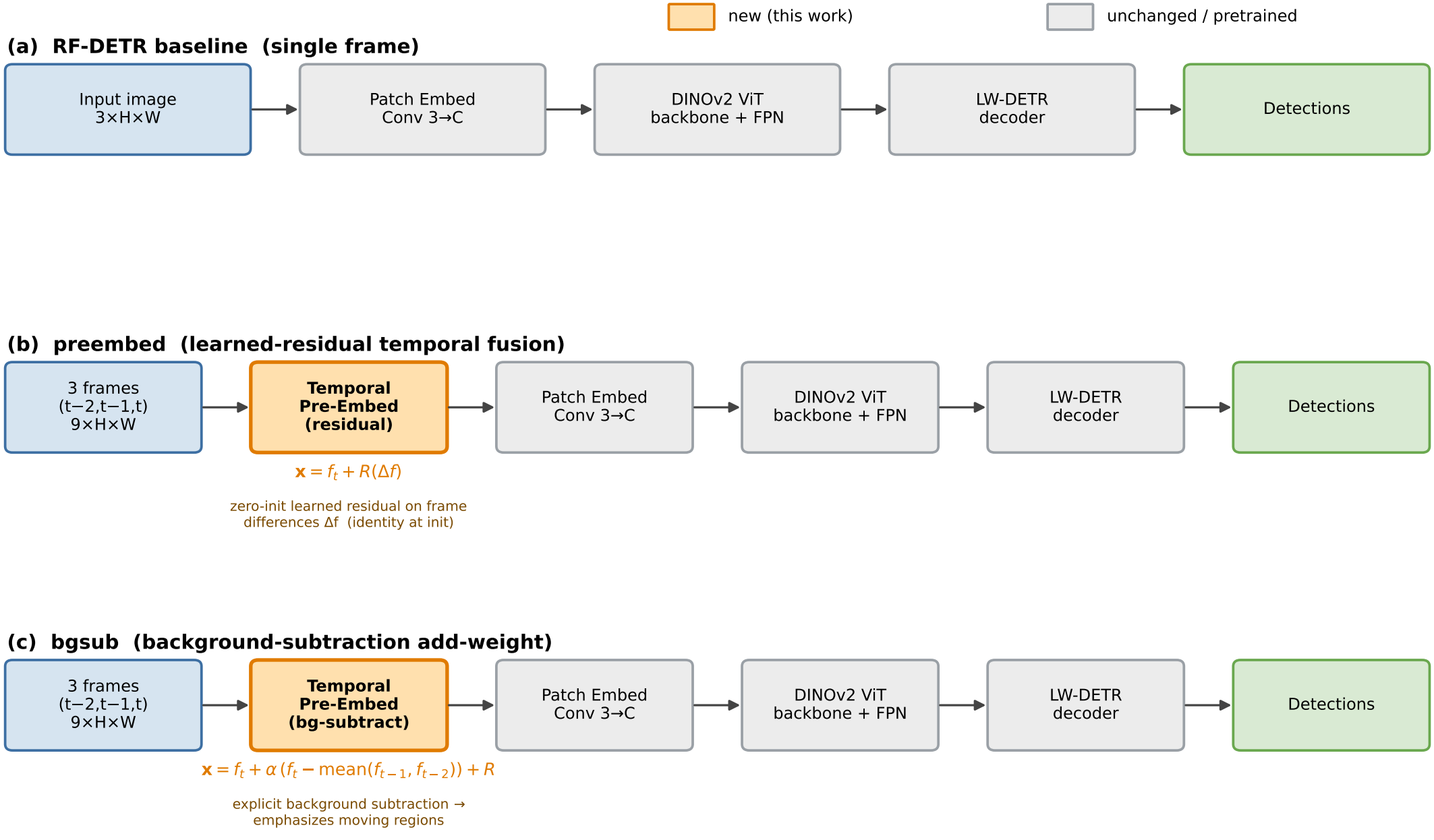
new (this work)
unchanged / pretrained
(a) RF-DETR baseline (single frame)
Input image
3×H×W
Patch Embed
Conv 3→C
DINOv2 ViT
backbone + FPN
LW-DETR
decoder
Detections
(b) preembed (learned-residual temporal fusion)
3 frames
(t−2,t−1,t)
9×H×W
Temporal
Pre-Embed
(residual)
Patch Embed
Conv 3→C
DINOv2 ViT
backbone + FPN
LW-DETR
decoder
Detections
$\mathbf{x} = f_t + R(\Delta f)$
zero-init learned residual on frame
differences Δf (identity at init)
(c) bgsub (background-subtraction add-weight)
3 frames
(t−2,t−1,t)
9×H×W
Temporal
Pre-Embed
(bg-subtract)
Patch Embed
Conv 3→C
DINOv2 ViT
backbone + FPN
LW-DETR
decoder
Detections
$\mathbf{x} = f_t + \alpha\,(f_t - \mathrm{mean}(f_{t-1}, f_{t-2})) + R$
explicit background subtraction →
emphasizes moving regions